

Register Number

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Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Computer Science and Engineering

Continuous Assessment Test – II

Regulations – R2021

Degree & Branch	B.E. & CSE				Semester	VI
Subject Code & Name	UCS2603 THEORY OF COMPUTATION					
Academic Year	2023-2024 EVEN	Batch	2021-2025	Date	25-04-2024	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Construct mathematical proofs related to computation and finite automata for a given language (K3)
CO2:	Identify relationship between finite automata and regular expressions (K3)
CO3:	Distinguish different types of grammars, analyse relationship between language and context free grammar and construct PDA for CFG (K4)
CO4:	Construct Turing machine for a given language (K3)
CO5:	Explain the decidability or undecidability of various problems and make use of it (K3)

Part – A ($4 \times 2 = 8$ Marks)

		KL	CO	PI
1.	Define Pushdown automata.	K1	CO3	1.4.1
2.	Construct Turing machine to find 1's complement of a binary number.	K3	CO4	2.1.3
3.	Explain the Move interpretation of a Turing machine with an example.	K2	CO4	2.1.3
4.	Classify the following languages under decidable and undecidable problems. - Regular language - Recursive language - CFL - Non Recursive Enumerable language - CSL - Recursive Enumerable language	K2	CO5	2.1.3

Part – B ($3 \times 6 = 18$ Marks)

		KL	CO	PI
5.	Construct a PDA for the following language. $L = \{0^n 1^{n+m} 2^m \mid m, n \geq 1\}$ Show the validation of $w = 011122$.	K3	CO3	1.4.1 2.4.1 13.4.1
6.	Show that the given language is not a Context Free Language (CFL). $L = \{x^n y^n z^n \mid n \geq 1\}$	K2	CO3	1.4.1 2.4.1
7.	Explain the closure properties of Recursive(R) and Recursively Enumerable (RE) languages with appropriate examples.	K2	CO5	1.3.1 1.4.1

Part – C (2 × 12 = 24 Marks)

		KL	CO	PI
8.	Construct a Chomsky Normal Form (CNF) representation for the given Context Free Grammar. $S \rightarrow ASB \mid \epsilon$ $A \rightarrow aAS \mid a$ $B \rightarrow SbS \mid A \mid bb$	K3	CO3	1.3.1 1.4.1 2.1.3 2.4.1 13.3.1
(Or)				
9.	Construct a Greibach Normal Form (GNF) representation for the given Context Free Grammar. $S \rightarrow AB$ $A \rightarrow BS \mid b$ $B \rightarrow SA \mid a$	K3	CO3	1.3.1 1.4.1 2.1.3 2.4.1 13.3.1
10.	Construct a Turing Machine for the following language. $L = \{0^n 1^n 2^n \mid n \geq 1\}$ Show the validation of $w = 001122$ and $w = 0012$	K3	CO4	1.3.1 1.4.1 2.1.3 2.4.1 13.3.1 13.4.1
(Or)				
11.	Construct a Turing Machine to subtract 2 numbers. $f(m, n) = m - n$ if $m > n$ $f(m, n) = B$ if $m \leq n$ Find the solution for 2-1 using the constructed Turing Machine.	K3	CO4	1.3.1 1.4.1 2.1.3 2.4.1 13.3.1 13.4.1